FOREST-AI

Conservation Monitoring System

A Comprehensive Guide for Conservation Officers and Field Administrators

Document: User Manual	**Version:** 1.0
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1. System Overview

FOREST-AI is a state-of-the-art conservation platform designed to monitor illegal deforestation, identify vulnerable areas, and optimize reforestation planning inside the Wilpattu Forest Reserve. By merging high-resolution Sentinel-2 satellite imagery composites with neural-network-driven analytics, the system supplies forest authorities with real-time, explainable telemetry to direct field patrols.

2. User Authentication & Session Security

To safeguard sensitive geographical alerts, FOREST-AI implements strict user authorization and access controls:

- **Access Lock:** All application pages require an active login session. Unauthenticated attempts are redirected back to the login screen.
- **Roles:** Administrators and Conservation Officers are supplied with unique profile descriptors (Username, Email, Role, Joined Date) that can be viewed by clicking the **Profile Dropdown** in the navbar.
- **Logout:** Click the Profile dropdown and choose 'Logout' to end your session safely, which clears all session cookies.

3. Main Dashboard & Layer Controls

The Dashboard serves as the central hub of the system, presenting a dynamic GIS map initialized on the Wilpattu boundary.

- **Map Layer Toggles:** The left sidebar contains checkboxes to overlay **Disturbances** (past canopy losses), **Risk Prediction** (modeled danger corridors), and **Reforestation** (ongoing replanting plots) on the map. You can combine layers to analyze spatial correlations.
- **Time Range Selection:** Filter the visual markers by adjusting the 'Start Year' dropdown and the 'End Year' range slider. The timeline spans from 2019 to the current year, updating instantly upon release.
- **Alert Center (Bell Icon):** Clicking the Bell in the top-right navbar expands a dropdown displaying the 4 most recent verified canopy losses. Each alert displays a colored severity level. Clicking any alert automatically pans the map and focuses on the target polygon.

4. Disturbance Detection (Vegetation Index Analysis)

The **Disturbance Detection** page lists and maps verified forest losses using automated pixel classification algorithms.

- **Disturbances Map:** Positioned above the table, this map highlights the precise boundaries of canopy loss. Hovering over a polygon shows its unique **Area ID** (e.g. Area 104) in a tooltip, and clicking reveals the affected extent and detection date.
- **NDVI Change:** The Normalized Difference Vegetation Index measures foliage density. The system flags localized vegetation declines (negative value changes). A larger negative score suggests complete clearance.
- **Confidence Scores:** Neural networks classify the likelihood that a detected anomaly represents actual deforestation. High confidence logs (>85%) should be treated as confirmed encroachment.

5. AI Risk Prediction & Explanations

The **Risk Prediction** page flags areas vulnerable to future clearance, using proximity metrics to assess threat levels.

- **Landscape Indicators:** Threat scores are calculated by measuring spatial proximity to logging corridors: distance to roads, distance to waterways, and distance to agricultural borders.
- **Explainable AI (XAI):** Every flagged risk area includes a human-readable explanation card. This explains what local factors contributed to the threat level (e.g. 'high risk due to a road corridor construction within 250m'), aiding resource allocation.

6. Reforestation Recommendations & Priority Matrix

The **Reforestation** module suggests optimal sites for canopy restoration, using a multi-criteria decision framework.

- **Suitability Index:** Classified as **Suitable**, **Moderately Suitable**, or **Unsuitable** by evaluating local water tables, soil moisture proxies, and past logging severity.
- **Priority Rankings:** Classified into **High**, **Medium**, or **Low** priority. High-priority sites represent highly degraded buffer zones that require immediate soil prep and tree planting.
- **Interactive Analytics:** The right sidebar renders live breakdown charts (Suitability Bar Chart and Priority Doughnut Chart) based on the latest field recommendations.

7. Monitoring Reports & PDF/CSV Exporter

The **Reports** page allows administrators to search database records, export raw telemetry, and generate print-ready reports.

- **Custom Date Filters:** Query the entire alerts database by entering a custom start date (defaults to 2019-01-01) and end date.
- **Generate Monthly Report:** Compiles a formal monthly conservation status document. This is optimized for A4 print sizes and removes screen-only UI elements when exported as a PDF.
- **CSV Export:** Downloads tabular database views into spreadsheet-compatible format for offline GIS integration.